The exact maximum growth factor for complete pivoting on real 5×5 matrices

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Abstract

Let $\rho_5^{\mathbb{R}}$ be the largest element growth in exact Gaussian elimination with complete pivoting on real 5×5 matrices, allowing every legal pivot tie. We prove $\rho_5^{\mathbb{R}} = \alpha = 4.132517078632472854223346853277\ldots$, the degree-61 algebraic lower bound of Chen, Edelman, and Urschel. Our contribution is the global upper bound. An inverse-elimination representation retains all entry bounds of the same matrix. A feasible contraction reduces the tail to two saturated geometries. A canonical representative theorem and a constrained algebraic maximum argument control the cross-saturated geometry; an attained six-variable fibre maximum reduces the negative-diagonal geometry to seventeen shared parameters. Exact rational certificates and feasible transports complete the source cover. Public certificates, verification code, and a partial Lean formalization accompany the proof.

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1. The problem and the result

Complete pivoting chooses a largest-modulus entry of each trailing Schur matrix. Although this bounds the elimination multipliers, successive updates can amplify entries. Sharp small-dimensional growth factors test how the constraints of an entire elimination path restrict that amplification. The exact real values in dimensions three and four are $9/4$ and 4 [4, 3]. Day and Peterson [5] and Gould [6] found numerical evidence for a fifth-pivot value near 4.1325. Chen, Edelman, and Urschel [2] identified a degree-61 algebraic value α, constructed an attaining matrix, and proved optimality within a prescribed equality family (their Theorem 2.2). Their global bound is 4.84 (Theorem 3.5). We establish the matching global upper bound in their Conjecture 2.1. The constant and attaining construction are theirs; the task here is to control every other feasible equality pattern.

Definition 1.1. For $0 \neq A \in \mathbb{R}^{n\times n}$ and a legal complete-pivoting path π, let $A_\pi^{(j)}$ denote the trailing matrix before pivot j is eliminated. Define

$$\rho(A,\pi) = \frac{\max_j \left\|A_\pi^{(j)}\right\|_{\max}}{\|A\|_{\max}}, \qquad \rho_n^{\mathbb{R}} = \sup_{A\neq 0} \sup_{\pi \text{ legal}} \rho(A,\pi).$$

A zero pivot makes the entire current trailing matrix zero; subsequent pivot magnitudes are defined to be zero.

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The norm $\|A\|_{\max}$ is the largest entry modulus. We use $\|\cdot\|_\infty$ for the maximum norm on vectors and its induced operator norm on matrices. Normalize $\|A\|_{\max} = 1$ and write p_j for the pivot magnitudes. Then

$$\rho(A, \pi) = \max_{1 \le j \le n} p_j. \tag{1}$$

Absorbing the path's row and column permutations into the original matrix reduces the proof to the fixed diagonal order. This treats every legal tied choice, without asserting that different tied paths coincide.

Theorem 1.2. *For exact complete-pivoting Gaussian elimination on real* 5×5 *matrices,*

$$\rho_5^{\mathbb{R}} = \alpha = 4.132517078632472854223346853277\ldots,$$

where α *is specified in Section 2. The bound holds for every legal pivot path, including ties and early termination.*

The result concerns a fixed dimension over $\mathbb{R}$, separately from asymptotic bounds such as [1, 8]; see also [5, 7] for growth and numerical error. The proof addresses two obstacles to passing from an attained candidate to a global bound. First, independently maximizing entries of different Schur complements need not produce any common input matrix. We therefore retain all initial and intermediate bounds in one inverse-elimination representation. Every analytic deformation acts on that same matrix. Second, a region containing an attaining matrix cannot be excluded under a rational hypothesis $F \ge \gamma < \alpha$. The certificates must separate strict exclusion from safety at the algebraic height α.

The argument has four stages. The first four pivot magnitudes are at most $4 < \alpha$, so only the fifth height F remains. A same-height contraction reduces its tail to a cross-saturated domain X or a proper negative-diagonal domain. The X bound combines rational exclusion with a canonical-representative theorem and a constrained maximum argument at the sharp height. For the negative-diagonal domain, three feasible paths give an attained maximum over six variables at fixed seventeen shared parameters; a finite, source-preserving cover then bounds that capacity. The dependency is canonical height, then complete X safety, then negative-diagonal safety.

A *physical source* means one reconstructed real matrix with its legal path and all its physical inequalities. A *framework* records the shared parameters of a fibre of feasible completions. A certificate box includes its inherited source conditions, not just coordinate endpoints. These distinctions are essential both to attainment and to composing a cover. The finite checks are universal implications on real regions, not numerical sampling.

This concise manuscript presents the proof chain, its acceptance rules, and a concrete certificate example. The accompanying *Full proof supplement* (FP) gives the detailed case analyses, finite exclusion systems, and exact data specifications. Its appendices contain the dependency map (A), canonical reduction details (B), transport data (C), source accounting (D), complete cross-saturated cover (E), and a full numerical certificate example (F). References prefixed "FP" use its own numbering. The separate *Supplementary Index*, S1–S8, locates the executable proof objects and their dependencies. Both documents form part of the material to be reviewed. Checking the large finite objects requires the supplied exact acceptors or an independent implementation of their stated rules.

2. The constant and an attained lower bound

Let

$$P_{61}(g) = \sum_{i=0}^{61} c_i g^i \tag{2}$$

be the polynomial in Chen, Edelman, and Urschel [2, Eq. (2.15)]. The Supplementary Index, S1, specifies its 62 integer coefficients and rational endpoints $L < U$, with $U - L = 10^{-200}$, isolating one real root. Set α equal to this root and $\alpha_- = L$. All coefficients agree exactly with the published polynomial. For

$$Q(t) = (1+t)^{61} P_{61}\left(\frac{4+5t}{1+t}\right),$$

the coefficients of degrees $0, \dots, 8$ are negative and those of degrees $9, \dots, 61$ are positive. Descartes' rule of signs therefore gives at most one positive root of Q, hence at most one root of P_{61} in $(4, 5)$. Exact evaluation gives $P_{61}(L) < 0 < P_{61}(U)$. Consequently α is the unique root in $(4, 5)$. The coefficient comparison, transformed coefficients, and endpoint signs are recorded in the Supplementary Index, S1. Every certificate comparison uses the full rational endpoints. We also use

$$\Lambda = \frac{1653}{400}, \qquad \gamma = \frac{4132517}{10^6}, \qquad 4 < \Lambda < \gamma < \alpha_- < \alpha. \tag{3}$$

An attaining completely pivoted matrix is established in [2, Theorem 2.2]; the exact identification above therefore gives $\rho_5^{\mathbb{R}} \geq \alpha$. S1 also records an independent reconstruction, with its nonzero-denominator and complete-pivoting checks. The rest of this paper proves the global upper bound.

3. An elementary reduction to the fifth pivot

We first bound all earlier pivots, including paths terminating before the fifth stage.

Lemma 3.1. *For a normalized real 3×3 matrix completely pivoted in the indicated order, let its pivot magnitudes be $1, t, g$. Then*

$$0 \leq t \leq 2, \qquad g \leq \Phi(t), \qquad \Phi(t) = \begin{cases} 2t, & 0 \leq t \leq 1, \\ t(3-t), & 1 \leq t \leq 2. \end{cases}$$

Each branch is sharp. In particular, $g \leq 9/4$.

Proof. The first Schur update is a difference of two quantities of magnitude at most one, so $t \leq 2$. If $t = 0$, its whole trailing block is zero. Otherwise the last update gives $g \leq 2t$, proving the first branch.

For $1 < t \leq 2$, normalize the first row and column to be nonnegative by independent sign changes. A positive entry of the first Schur complement is at most one, so the second pivot is negative. Write the tail as

$$\begin{pmatrix} -t & \gamma_0 \\ \eta_0 & \zeta_0 \end{pmatrix}.$$

Let the relevant first-row and first-column coordinates be $x, u, y, v \in [0, 1]$, and set

$$c = t - 1, \quad s = xv, \quad z = uy, \quad r = uv, \quad w = xy.$$

The original entry at the pivot is $w - t$, whence $w \geq c$. Furthermore $sz = wr$ and $r \geq sz$. Intersecting the original entry bounds with the Schur pivot bounds gives

$$\begin{aligned} \gamma_0 &\in [-1 - \min(c, s),\ 1 - s], \\ \eta_0 &\in [-1 - \min(c, z),\ 1 - z], \\ \zeta_0 &\in [-1 - \min(c, r),\ 1 - r]. \end{aligned} \tag{4}$$

We claim that, with $M = t^2(3-t)$,

$$t(1-r) + (1+\min(c,s))(1+\min(c,z)) \le M, \tag{5}$$
$$t(1+\min(c,r)) + (1+\min(c,s))(1-z) \le M. \tag{6}$$

For (5), replace r by sz. When $s, z \le c$ the resulting function increases in either variable, since its corresponding derivative is $1-cz$ or $1-cs$; its maximum is at $s = z = c$. If $s \le c \le z$, it is $t[2+s(1-z)] \le t[2+c(1-c)] = M$. The other mixed case is symmetric. If $s, z \ge c$, the bound is $t(1-c^2)+t^2 = M$.

For (6), the shared-product relations imply $z \ge cr$, and also $z \ge r$ whenever $s \le c$. The latter includes $s = 0$: then $sz = wr$ and $w \ge c > 0$ force $r = 0$. Thus no division by s is needed. Separate four closed cases. If $r \ge c, s \ge c$, then $z \ge cr \ge c^2$, and the expression is at most $t^2 + t(1-c^2) = M$. If $r \ge c, s \le c$, then $z \ge r \ge c$, and the bound $t^2 + t(1-c) = 2t \le M$ suffices. If $r \le c, s \le c$, the expression is at most

$$t(1+r) + (1+s)(1-r) \le 2t \le M;$$

the first difference is $(c-s)(1-r) \ge 0$, and $M - 2t = tc(1-c) \ge 0$. Finally, if $r \le c, s \ge c$, then $r - z \le r(1-c) \le c(1-c)$. The expression is at most $t[2+c(1-c)] = M$. Interchanging s, z gives the symmetric form of (6).

Put $A_s = 1 + \min(c,s)$, $B_s = 1 - s$, and similarly for z, r. By (4),

$$t\zeta_0 + \gamma_0\eta_0 \le tB_r + \max(A_sA_z, B_sB_z) \le M.$$

The first product uses (5); the second is at most $t + 1 \le M$. On the other side,

$$-(t\zeta_0 + \gamma_0\eta_0) \le tA_r + \max(A_sB_z, B_sA_z) \le M$$

by (6) and its symmetric version. Thus the magnitude of the tail determinant is at most M, and division by t proves $g \le t(3-t)$.

Sharp examples are

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & t & t \\ 0 & -t & t \end{pmatrix} \quad (0 \le t \le 1), \qquad \begin{pmatrix} 1 & 1 & c \\ 1 & -c & -1 \\ c & -1 & 1 \end{pmatrix}, \quad c = t-1 \quad (1 \le t \le 2).$$

Finally $9/4 - t(3-t) = (t-3/2)^2$. □

Corollary 3.2. *The first four pivot magnitudes of a normalized real completely pivoted matrix of order at most five satisfy*

$$p_1 = 1, \qquad p_2 \le 2, \qquad p_3 \le 9/4, \qquad p_4 \le 4.$$

Consequently $\rho(A, \pi) > \alpha$ implies $p_5 > \alpha$.

Proof. Any leading submatrix in the chosen pivot order is completely pivoted, and a Schur tail divided by its first pivot magnitude is normalized and completely pivoted. Put $x = p_2$, $y = p_3/p_2$ when these denominators are nonzero. Lemma 3.1 gives

$$xy \le \Phi(x), \qquad p_4 \le x\Phi(y).$$

If $x \le 1$, the second bound is at most $9/4$. If $1 \le x \le 3/2$, it is at most $27/8$. If $3/2 \le x \le 2$, then $y \le 3 - x \le 3/2$. The function Φ is nondecreasing on $[0, 3/2]$, so

$$p_4 \le x\Phi(3-x) = x^2(3-x) \le 4,$$

using $4 - x^2(3-x) = (2-x)^2(x+1)$. Zero pivots imply zero subsequent growth. Equation (1) and $\alpha > 4$ finish the proof. □

4. A complete two-step lift

Undoing the first two elimination steps retains the original matrix and couples its entry bounds with those of both Schur tails. All subsequent variables describe this one matrix. Let $u, x, v, q \in \mathbb{R}^3$, $p, k > 0$, and $e, \beta \in \mathbb{R}$. Set

$$M = \begin{pmatrix} 1 & -e & v^T \\ \beta & p - e\beta & (q + \beta v)^T \\ u & px - eu & D + xq^T + uv^T \end{pmatrix}, \qquad D = kC, \quad C_{00} = 1. \tag{7}$$

Subscripts on 3×3 blocks run from zero to two. Its first two Schur tails are

$$M^{(2)} = \begin{pmatrix} p & q^T \\ px & D + xq^T \end{pmatrix}, \qquad M^{(3)} = D. \tag{8}$$

Proposition 4.1. *Suppose C is completely pivoted in the fixed order, with $\|C\|_{\max} = 1$. Then (7) is normalized and completely pivoted, with first three pivot magnitudes $1, p, k$, if and only if*

$$|e|, |\beta|, \|u\|_\infty, \|v\|_\infty, \|x\|_\infty \le 1, \tag{9}$$

$$|p - e\beta|, \ \|px - eu\|_\infty, \ \|q + \beta v\|_\infty \le 1, \tag{10}$$

$$\|q\|_\infty \le p, \qquad \left\|D + xq^T\right\|_{\max} \le p, \tag{11}$$

$$\left\|D + xq^T + uv^T\right\|_{\max} \le 1. \tag{12}$$

Every normalized fixed-order completely pivoted matrix with three nonzero leading pivots has this representation after row-sign normalization of those pivots.

Proof. The identities (8) show sufficiency: the first matrix has entries bounded by one, the first tail by p, and the second tail is kC. Conversely recover e, β, u, v from the initial matrix, then p, x, q from its first Schur tail. The remaining tail is uniquely D; division by its positive leading pivot gives C. The inequalities are precisely the initial and first-stage bounds. No independent completion is chosen for different inequalities. □

The alternative notation $q = py$ makes the bottom block $kC + uv^T + pxy^T$. The actual variable q, rather than y, will be used in the seventeen-dimensional reduction. Write

$$C = \begin{pmatrix} 1 & a & b \\ c & ca + \rho & cb + \sigma \\ d & da + \tau & db + \omega \end{pmatrix}. \tag{13}$$

The fixed-order core conditions are $\|C\|_{\max} \le 1$ and $|\sigma|, |\tau|, |\omega| \le |\rho|$. For $\rho \neq 0$, set

$$h(C) = \omega - \tau\sigma/\rho, \qquad F = k|h(C)|.$$

If $\rho = 0$, the final tail is zero; set $h(C) = F = 0$. Lemma 3.1 also yields

$$k \le 9/4, \quad |h(C)| \le 9/4, \quad k \le 2p, \quad |h(C)| \le 2|\rho|. \tag{14}$$

All these quantities belong to the same source.

5. Constructive saturation of the last tail

Contractions of actual rows and columns produce a representative with the same last-pivot height and one of two saturated tails. This is an existence statement; it does not assume that an arbitrary maximizer already has the candidate's tight constraints.

Lemma 5.1 (Same-height diagonal saturation)**.** *Every complete lift* (7) *with* $F > 0$ *has another complete lift with the same* p, k, F *whose normalized tail satisfies either*

$$D: \quad |\omega| = |\rho|, \quad |\sigma|, |\tau| \le |\rho|,$$

or

$$X: \quad |\sigma| = |\tau| = |\rho|, \quad |\omega| \le |\rho|.$$

The construction is valid on every legal tied pivot path.

Proof. For $0 \le \lambda, \mu \le 1$, let $R_\lambda = \operatorname{diag}(1, \lambda, 1)$, $K_\mu = \operatorname{diag}(1, \mu, 1)$. Replace

$$C \mapsto R_\lambda C K_\mu, \quad (u, x) \mapsto (R_\lambda u, R_\lambda x), \quad (v, q) \mapsto (K_\mu v, K_\mu q).$$

The new full matrix is $\operatorname{diag}(I_2, R_\lambda) M \operatorname{diag}(I_2, K_\mu)$. Its initial and first-tail entries are only contracted in magnitude, while the pivots $1, p, k$ do not change. The core tail transforms as

$$(\rho, \sigma, \tau, \omega) \mapsto (\lambda\mu\rho, \lambda\sigma, \mu\tau, \omega). \tag{15}$$

Since $F > 0$, $\rho \neq 0$. Put

$$S_0 = |\sigma/\rho|, \quad T_0 = |\tau/\rho|, \quad W_0 = |\omega/\rho|, \qquad S_0, T_0, W_0 \in [0, 1].$$

If $W_0 \ge S_0 T_0$, choose $\lambda = \max(T_0, W_0)$, $\mu = W_0/\lambda$. In this case $W_0 = 0$ would force $S_0 T_0 = 0$, hence $\omega = 0$ and $\sigma\tau = 0$. This would give $F = k|\omega - \tau\sigma/\rho| = 0$, contrary to the hypothesis. Thus $W_0 > 0$, the division is valid, and $\lambda\mu > 0$. Now $\lambda \ge T_0$, $\mu \ge S_0$, and $\lambda\mu = W_0$, so the new completely pivoted tail lies on D. If $W_0 \le S_0 T_0$, take $\lambda = T_0, \mu = S_0$. Here $S_0 T_0 = 0$ would also force $W_0 = 0$ and $F = 0$, so $\lambda\mu = S_0 T_0 > 0$; the new tail lies on X. At the positive-product junction $W_0 = S_0 T_0 > 0$, the constructions agree. Finally,

$$\omega - \frac{(\mu\tau)(\lambda\sigma)}{\lambda\mu\rho} = \omega - \frac{\tau\sigma}{\rho}.$$

This preserves F and all the complete early-stage conditions. No exchange of unequal pivots is introduced. □

Zero fifth pivots are already safe by Corollary 3.2, even when the fourth-stage tail is nonzero of rank one. A zero fourth pivot, in contrast, forces that whole tail to vanish. No extension of the lemma to zero targets or zero-product junctions is needed.

Actual sign changes on the final rows and columns reduce every X-tail to

$$H = \begin{pmatrix} r & r \\ r & w \end{pmatrix}, \qquad r > 0, \quad |w| \le r, \quad F = r - w. \tag{16}$$

Here H is the *actual* fourth-stage tail, whose entries are k times the normalized core-tail entries. Indeed, for $\left(\begin{smallmatrix} \epsilon_1 r & \epsilon_2 r \\ \epsilon_3 r & W \end{smallmatrix}\right)$, multiply rows by (ϵ_1, ϵ_3) and columns by $(1, \epsilon_1\epsilon_2)$. This gives (16), with $w = \epsilon_1\epsilon_2\epsilon_3 W$, and preserves all absolute matrix bounds. Similarly every D-tail has a form

$$H = \begin{pmatrix} r & s \\ t & \chi r \end{pmatrix}, \quad r > 0, \quad 0 \le s, t \le r, \quad \chi \in \{-1, 1\}. \tag{17}$$

If $\chi = 1$, its height is $r - st/r \le r \le 4$; a zero arm also gives height $r \le 4$. If $s = r$ or $t = r$, exchanging the corresponding tied final columns or rows and normalizing signs gives (16) at the same height. The sole additional domain is therefore

$$H = \begin{pmatrix} r & s \\ t & -r \end{pmatrix}, \qquad 0 < s, t < r, \qquad F = r + \frac{st}{r}. \tag{18}$$

We call this the proper negative diagonal domain.

6. The complete cross-saturated bound

For (16), the actual third Schur block is

$$D = \begin{pmatrix} k & A & B \\ ck & r + cA & r + cB \\ dk & r + dA & w + dB \end{pmatrix}. \tag{19}$$

The complete lift retains (9)–(12) and $\|D\|_{\max} \le k$, with no rank or cofactor-sign assumption.

Proposition 6.1 (Complete cross-saturated bound)**.** *Every complete lift with* (19)*, $p, k, r > 0$, and $|w| \le r$ satisfies $F = r - w \le \alpha$.*

Its proof has two different parts: an exact-height argument in a qualified positive chamber, followed by a complete source cover which also includes every other sign and degenerate boundary. We give the analytic chain first. The full entrywise derivations and finite source systems are retained in the full proof supplement.

6.1. Complete canonical sources and prescribed heights

Write $u = (U, V, T)^T$, $x = (X, Y, Z)^T$, and define

$$\begin{gathered} \delta = U - \beta X, \quad \Gamma = V - \beta Y, \quad \nu = \beta Z - T, \\ \tau = UY - XV, \quad \Delta = UZ - TX, \quad \mu = VZ - TY. \end{gathered} \tag{20}$$

The cofactor Γ is distinct from the threshold γ.

Definition 6.2 (Canonical source)**.** The chamber $\mathcal{K}$ consists of complete lifts with $e = U = 1$, $0 < V < 1$, $0 < \beta, T, X, Y, Z \le 1$, all six cofactors positive, and the following simultaneous representation:

$$\begin{gathered} u = (1, V, T)^T, \quad x = (X, Y, Z)^T, \quad \widehat{v} = (-g, h, v)^T, \quad \widehat{q} = (-j, -1 - \beta h, q)^T, \\ D = \begin{pmatrix} k & -a & -b \\ tk & r - ta & r - tb \\ -Ek & r + Ea & w + Eb \end{pmatrix}, \qquad S = D + x\widehat{q}^T, \quad O = S + u\widehat{v}^T, \\ P = \beta\widehat{v} + \widehat{q}, \quad L = px - u, \qquad M = \begin{pmatrix} 1 & -1 & \widehat{v}^T \\ \beta & p - \beta & P^T \\ u & L & O \end{pmatrix}. \end{gathered} \tag{21}$$

Here $p, k, r > 0$, $|t| \le 1$, $0 \le E \le 1$, $0 \le a, b \le k$, $|w| \le r$, and every bound in Proposition 4.1, together with $\|D\|_{\max} \le k$, holds. In particular $P_1 = -1$. No sign of g, h, v, j, q, tail contact, or equality of row scales is assumed. The final-pivot height is $F = r - w$.

Set $H_0 = 33/8$, $\Lambda = 1653/400$, and fix the head $\eta = (t, \beta, V, T, X, Y, Z)$. The physical bands at $F > 4$ give the following entrance inequalities:

$$\begin{aligned} &2 < r \le k < 3, \quad 1 < p < 1+\beta < 2, \quad g, j, h, q > 0, \\ &V < t \le 1, \quad X > 1/(1+\beta) > \delta, \quad \Delta > \tau, \quad Ek < 1 - T, \\ &Q_o := t\delta - \Gamma > 0, \quad \Omega := t\Delta + E\tau - \mu > 0, \quad \zeta := t\nu - E\Gamma > 0. \end{aligned} \tag{22}$$

At $F \ge H_0$, also $v > 0$ and $\ell := \nu - E\delta > 0$. These are consequences of a complete source, not extra restrictions on the original problem. Their derivation, including thirteen fixed safe-tail positivity checks and four fixed univariate checks, is in FP, Lemma B.2–FP, Lemma B.10.

The head inequality $pZ - T \le 1$ permits the normalization $p = \bar{p} := (1+T)/Z$. Indeed increasing p to $\bar{p}$ relaxes the receiver and stage bands; the changing head entries obey $p'X < 2$, $p'Y < 1+V$, and $p'Z \le 1+T$, while $p'x_i - u_i \ge -u_i \ge -1$. Here $1+V-(1+\beta)Y = 1-Y+\Gamma > 0$ and $\bar{p} < 1+\beta$, which also gives $0 < p' - \beta < 1$. Thus every intermediate matrix is physical, and the other coordinates and the height are fixed.

Lemma 6.3 (Complete first- and middle-column capacities)**.** *Fix head coordinates and E satisfying the head inequalities in* (22)*, with all six cofactors positive, $0 \le E \le 1$, and put $p = \bar{p}$, where $1 < \bar{p} < 1+\beta < 2$. Define three lower and four upper affine functions by*

$$\begin{aligned} (\lambda_i, \eta_i)_{i=1}^3 &= \left((1/X, p/X), (t/Y, p/Y), (t/Y, (1+V)/Y)\right), \\ (\xi_1, \kappa_1) &= \left(\frac{1+\beta}{\delta}, \frac{\beta}{\delta}\right), \quad (\xi_2, \kappa_2) = \left(\frac{1+T}{\Delta}, \frac{E+T}{\Delta}\right), \\ (\xi_3, \kappa_3) &= \left(\frac{V+\beta}{\Gamma}, \frac{\beta t}{\Gamma}\right), \quad (\xi_4, \kappa_4) = \left(\frac{V+T}{\mu}, \frac{VE+Tt}{\mu}\right). \end{aligned}$$

Then, for $k > 2$, a complete first column exists if and only if

$$k \le K(E) := \min_{1 \le i \le 3,\, 1 \le j \le 4} K_{ij}, \qquad K_{ij} = \frac{\eta_i + \xi_j}{\lambda_i + \kappa_j}. \tag{23}$$

One witness is

$$j = \max_i (\lambda_i k - \eta_i), \qquad g = \max\{k - 1 - Xj, (tk - 1 - Yj)/V\}. \tag{24}$$

If $K > 2$, supply that first column at $k = K$. Put $H = (p-1)/\beta$ and

$$\begin{aligned} A_1 &= 1 + Y + t(1 - X), & B_1 &= Q_o, \\ A_2 &= \frac{t(1+Z) + E(1+Y)}{t+E}, & B_2 &= \frac{\zeta}{t+E}, \\ C_1 &= 1 + Y + t(p - X), & D_1 &= \Gamma + t\beta X, \\ C_2 &= \frac{tK + E(1+Y)}{t+E}, & D_2 &= \frac{E\Gamma}{t+E}. \end{aligned}$$

The middle column carries a prescribed $r > 2$ exactly when

$$r \le R_m := \min \left\{ \begin{array}{c} C_1, C_2, A_1 + B_1 H, A_2 + B_2 H, \\ \dfrac{B_i C_j + D_j A_i}{B_i + D_j} \quad (i, j = 1, 2) \end{array} \right\}. \tag{25}$$

Its witness at the same p, K, E is

$$h = \max\{0, (r - A_1)/B_1, (r - A_2)/B_2\}, \qquad a = \frac{r - 1 - Y + \Gamma h}{t}. \tag{26}$$

It has $O_{11} = 1$, and $O_{01} = -1$ or $O_{21} = 1$.

Proof. For the first column write $j_L = \max_i(\lambda_i k - \eta_i)$ and $j_U = \min_j(\xi_j - \kappa_j k)$. The stage bounds impose the first two lower bounds on j. The constraints on g are

$$\max\{k-1-Xj, (tk-1-Yj)/V\} \le g \le \min\{1, (1-j)/\beta, (1-Ek-Zj)/T\}.$$

Comparing their two lower bounds with the last two upper bounds gives the four terms in j_U; comparison with $g \le 1$ gives the third lower bound in j_L. Thus feasibility is exactly the twelve comparisons $j_L \le j_U$, equivalent to (23). All denominators are positive. The proposed witness satisfies $0 < j < 1$ and $0 < k-2 < g \le 1$. The remaining lower bands follow from $O_{00} > k-2 > 0$, $O_{10} > t(k-g)-1 > -1$, and $-1 \le O_{20} < 0$; the corresponding stage entries are obtained by adding nonnegative $u_i g$, and their required upper bounds were included above. In particular the complete column gives $K < 3$ and $EK < 1-T$. Raising an existing k to K using this witness fixes the other columns and relaxes their core bounds, so it preserves the source height.

For the middle column, lowering a to $(r-1-Y+\Gamma h)/t$ sets $O_{11} = 1$ and preserves its bands: $O_{01} < \delta - X < 0$, while the affected bottom entries remain positive. Substitution gives the four exact tests

$$\begin{aligned} t(1+O_{01}) &= A_1 + B_1 h - r, \\ t(1-O_{21}) &= (t+E)(A_2 + B_2 h - r), \\ t(p+S_{01}) &= C_1 - D_1 h - r, \\ t(K-r-Ea) &= (t+E)(C_2 - D_2 h - r). \end{aligned}$$

Together with $0 \le h \le H$, these are two increasing and two nonincreasing affine bounds on r; their eight comparisons give (25). Since $B_1, B_2, D_1 > 0$, $D_2 \ge 0$, every denominator remains positive also at $E = 0$. The smallest feasible h is (26); $A_2 \le 2 < r$ implies $h > 0$ and an active increasing bound. The unused bands follow from $0 < a < p < 2 < K$, $D_{11} = 1+Y-\Gamma h \in (0,2)$, $S_{11} = 1 - Vh \in (0,1]$, and $0 < r - p \le S_{21} \le O_{21} \le 1$. Finally $0 < h \le H < 1$, $1 + \beta h \le p$, and $P_1 = -1$ supply the receiver and prefix bounds. This supplies two columns of the same matrix, not separate optimizers. □

For tail recovery retain these head inequalities and $0 \le E < \min\{1, \nu/\delta\}$, with $K, R_m > 2$. Write $y = -w$ and set

$$\begin{gathered} b_0 = 1 + X, \quad a_T = 1 - Y + \Gamma b_0/\delta, \quad c_T = 1 + Z + \nu b_0/\delta, \\ R_T(b) = a_T + Q_o b/\delta, \quad C_T(b) = c_T - \ell b/\delta, \quad C_0 = 1 + Z + E b_0, \\ A_c = \mu + \Delta + \tau, \quad B_c = \Delta + \nu + \delta, \quad \mathcal{C} = A_c + \Omega K, \quad \mathcal{D} = \ell a_T + Q_o c_T. \end{gathered} \tag{27}$$

The quantities R_T, C_T are scalar capacities. Expansion in (21) gives the same-matrix slack identities

$$\begin{aligned} J_a &:= \mu(1+O_{02}) + \Delta(1-O_{12}) + \tau(1+O_{22}) \\ &= A_c + \Omega b - \Delta r - \tau y, \\ J_b &:= \Delta(1-P_2) + \nu(1+O_{02}) + \delta(1+O_{22}) \\ &= B_c - \ell b - \delta y, \\ \ell J_a + \Omega J_b &= \Delta(\mathcal{D} - \ell r - Q_o y), \\ C_0 - y &= (1+O_{22}) + E(1+O_{02}) + (EX+Z)(1-P_2) + \ell v. \end{aligned} \tag{28}$$

Every weight is nonnegative, and $\ell v > 0$ at high value. Since $b \le K$, every normalized high source therefore maps to

$$\begin{aligned} 0 \le y \le r \le R_m, \qquad & y \le C_0, \\ \Delta r + \tau y \le \mathcal{C}, \qquad & \ell r + Q_o y \le \mathcal{D}. \end{aligned} \tag{29}$$

Here $y = F - r > 0$ because the source bounds give $r < 5/2 < F$. No tail contact was used to derive this polygon.

Lemma 6.4 (Physical recovery of the relevant polygon points)**.** *At the fixed head and E, every wall point $y = r \geq H_0/2$ of (29) is realized by a complete source. If the polygon maximum of $r + y$ exceeds H_0, that maximum also has a complete realization. No claim is made for arbitrary low-value polygon points.*

Proof. First supply the first and middle columns by Lemma 6.3 at common p, K, E, r. For a wall point with $r \leq R_T(K)$, take

$$b = \max\{b_0, \delta(r - a_T)/Q_o\}, \quad v = (b - b_0)/\delta, \quad q = 1 - \beta v, \quad w = -r. \tag{30}$$

The polygon gives $b_0 \leq b \leq K$, $R_T(b) \geq r$, and $C_T(b) \geq r$. Also $K \leq b_0 + \delta(p - 1)$, one of the twelve first capacities, gives $0 \leq v \leq p - 1 < 1$ and $0 < q \leq 1$. Thus $P_2 = 1$, $O_{02} = -1$, $O_{12} = 1 + r - R_T(b) \leq 1$, and $O_{22} = -r + C_T(b) - 1 \geq -1$. The opposite bounds follow from $O_{12} > r - K > -1$ and $O_{22} < 2 - r < 0$, using $EK < 1 - T$. The stage lower bounds are $S_{02} = -1 - v \geq -p$, $S_{12} > -1$, and $S_{22} \geq -1 - Tv \geq -p$; their upper bounds follow from $S_{i2} = O_{i2} - u_i v \leq 1 < p$. For the core, $r - tb \geq r - K > -K$ and $-r + Eb \geq -r \geq -K$; the upper bounds follow by subtracting the nonnegative completion terms from O_{12}, O_{22}.

If $r > R_T(K)$, instead take

$$b = K, \quad q = \frac{1 + V + (t - V)K - r}{\tau}, \quad v = \frac{Xr - (tX - Y)K - X - Y}{\tau}, \quad w = -r. \tag{31}$$

Then $O_{02} = -1$, $O_{12} = 1$; the first oblique polygon bound is exactly $O_{22} \geq -1$. These formulas join (30) at $r = R_T(K)$, after which $v > 0$ and $P_2 < 1$. The needed upper bound on v follows independently from

$$\begin{aligned} \mu(p + S_{02}) = {} & X(V + T)(r - 2) + \mu(p - S_{00}) \\ & + XV[(1 + O_{20}) + (1 + O_{22})] \\ & + XT[(1 - O_{10}) + (1 - O_{12})] > 0. \end{aligned}$$

The right side involves only bounds already verified. Hence $v < p - 1 < 1$, and $q = (K - 1 - v)/X > 0$, yielding $0 < P_2 < 1$, $q < 1$. All remaining tail bands follow as in the first case. This proves recovery on the wall with all three columns at the same p, K, E, r.

Now take a polygon maximizer with height greater than H_0. Its reconstructed first and middle columns give $r \leq \min\{3 - \beta, 2 + \beta - T\} < 5/2$, so $y > 0$. A nonwall maximizer with $y = C_0$ would be realized by $b = b_0, v = 0, q = 1$, contradicting the high-source inequality $v > 0$. At any other nonwall maximizer the second oblique bound is active: otherwise the direction $(-\tau, \Delta)$ preserves the first and increases height by $\Delta - \tau > 0$. Use the identities $\Delta Q_o - \tau \ell = \Omega \delta$ and $\Delta \mathcal{D} = \ell A_c + \Omega B_c$ and set

$$b = (B_c - \delta y)/\ell, \qquad v = (b - b_0)/\delta, \qquad q = 1 - \beta v, \qquad w = -y.$$

They give $b_0 < b \leq K$, $R_T(b) = r$, $C_T(b) = y$. Thus $P_2 = 1$ and the three tail contacts are $(-1, 1, -1)$. The preceding entrywise bounds apply, with the last core bound now $-y + Eb \geq -y \geq -r \geq -K$. This realizes the maximum. □

Proposition 6.5 (Prescribed-target representative)**.** *Fix the head $\eta = (t, \beta, V, T, X, Y, Z)$. If a source in $\mathcal{K}$ has height at least f, where $\Lambda \leq f < 5$, a complete source at the same head has height exactly f and the contacts*

$$\begin{gathered} w = -r, \quad r = f/2, \quad pZ = 1 + T, \quad P_1 = -1, \quad P_2 = 1, \\ O_{10} = O_{11} = 1, \quad (O_{02}, O_{12}, O_{22}) = (-1, 1, -1), \\ S_{10} = p, \quad O_{20} = -1, \quad S_{00} < p, \quad P_0 > -1. \end{gathered} \tag{32}$$

These are contacts of one complete matrix; they need not hold at the original source.

Reduction argument. The preceding two lemmas identify the actual high maxima with polygon maxima at each admissible E. The high fixed-head fibre is compact: restricting to $F \geq f$ gives $r \geq f/2$, $k \geq r$, $p \geq f/4$, while all other coordinates are bounded by the physical bands. At a hypothetical nonwall maximum of $r + y$, the active right derivatives of the finite polygon budgets force one of eight specified stationary contact systems. Their exact contradiction certificates exclude that obstruction. This proves that a same-head maximum has a wall representative, including capacity ties and $E = 0$; the full active-branch calculation and the eight systems are in FP, Theorem B.16 and FP, Proposition B.15. Lowering both wall coordinates preserves every inequality in (29); Lemma 6.4 reconstructs height exactly f.

Choose such a wall minimizing E. At this wall the high-value entrance bounds give $Eb > r - 2$ and $Ek < 1 - T$, hence $0 < E < 1/2$. Decreasing E preserves first- and middle-column feasibility: their affected bottom entries move toward zero, and every first-column capacity in (23) is nonincreasing in E. Thus a tail capacity must be active; its positive slack identity (28) forces the three original tail contacts and $P_2 = 1$ or $b = k$. At this completed high-value source, the common column witnesses at the full capacity $K(E)$, and the strict-gap result for that same matrix, give a representative in $\mathcal{R}_* \subset \mathcal{K}$. Its additional contacts are precisely

$$\begin{gathered} w = -r, \quad pZ = 1 + T, \quad O_{10} = O_{11} = 1, \quad (O_{02}, O_{12}, O_{22}) = (-1, 1, -1), \\ (S_{00} = p \text{ or } S_{10} = p), \quad (P_0 = -1 \text{ or } O_{20} = -1), \\ (O_{01} = -1 \text{ or } O_{21} = 1), \quad (P_2 = 1 \text{ or } b = k). \end{gathered} \tag{33}$$

All contacts refer to the same matrix in (21). The physical bands and cofactor conditions of $\mathcal{K}$ remain in force; the displayed contacts alone do not define a complete source.

The three finite inputs have distinct roles. The eight stationary systems used above certify derivative positivity at $F \geq H_0$, under the full nonwall contact hypotheses of FP, Proposition B.15; they do not exclude arbitrary nonwall sources. The eight head-difference systems in FP, Proposition B.17, after the same-head wall reduction, prove $tX - Y \leq 0 \Longrightarrow F \leq H_0$ for complete sources in $\mathcal{K}$. Hence our target $f \geq \Lambda > H_0$ has $tX - Y > 0$. The last six systems give the implication

$$M \in \mathcal{R}_*, \quad (S_{00} = p \text{ or } P_0 = -1) \quad \Longrightarrow \quad F < \Lambda.$$

To see the finite coverage, combine each exceptional contact with the remaining first- and middle-column alternatives:

Case	First-column contact	Stage contact	Middle contact
1	$P_0 = -1$	$S_{00} = p$	$O_{01} = -1$
2	$P_0 = -1$	$S_{00} = p$	$O_{21} = 1$
3	$P_0 = -1$	$S_{10} = p$	$O_{01} = -1$
4	$P_0 = -1$	$S_{10} = p$	$O_{21} = 1$
5	$O_{20} = -1$	$S_{00} = p$	$O_{01} = -1$
6	$O_{20} = -1$	$S_{00} = p$	$O_{21} = 1$

The rows need not be disjoint: ties belong to every applicable row. In all six systems, the last alternative in (33) is retained as $(1 - P_2)(k - b) = 0$, with both factors nonnegative. The excluded height hypothesis is the exact integer inequality $800r - 1653 \geq 0$, since $F = 2r$ on this wall. It is not replaced by the lower threshold $F \geq H_0$.

For each case, the source identities and physical bands imply a closed necessary polynomial system on a rational root box; the source-to-box inclusion and necessary-row derivations are given in FP, Proposition B.19. Every auxiliary retains its defining equation, and closing an interval enlarges the necessary domain. The exact acceptor verifies every terminal contradiction and every covered split, so finite induction proves all six roots empty. Therefore no

exceptional representative with $F \geq \Lambda$ remains. None of these three families uses the later canonical-height or complete-X conclusion. We have thus obtained $S_{00} < p$ and $P_0 > -1$; the remaining first-column capacity is strictly selected:

$$k = K_{24} = \frac{p\mu + Y(V+T)}{V(tZ+EY)}, \qquad k < K_{13}, K_{14}, K_{23}.$$

If $P_2 < 1$, then $b = k$. The strict-gap lemma for a common completed tail (FP, Lemma B.7) excludes the other eight first-column capacities. This exclusion requires a completed high-value source; four selected capacities are not an independent relaxation. Together with the three strict comparisons above, it makes the full $K(E)$ locally equal to $K_{24}(E)$. The only active tail capacity has numerator containing ΩK_{24}, and

$$\frac{d(\Omega K_{24})}{dE} = -\frac{[p\mu + Y(V+T)]V\mu(tX-Y)}{[V(tZ+EY)]^2} < 0.$$

Decreasing E would improve that capacity and preserve the other strict gaps, contradicting minimality. Hence $P_2 = 1$, and the common reconstruction gives (32). The remaining active-branch and boundary case analyses are in FP, Proposition B.21. □

6.2. Simultaneous reconstruction and localization

A target representative is a complete source of height $\Lambda \leq F < 5$, with (32) and the proved high-value qualifications in FP, (B.31). Write $p = 1 + s$, $r = 1 + R$, $F = 2(1 + R)$, and use $z = (\beta, V, s, Y, Z, v, h, b, R)$. Define

$$\begin{gathered} T = (1+s)Z - 1, \quad q = 1 - \beta v, \quad \Gamma = V - \beta Y, \quad \nu = \beta Z - T, \quad \mu = VZ - TY, \\ A = R + Y + \Gamma v, \quad C = R - Z + \nu v, \quad W = R(Y+Z) + \mu v, \\ N = V(Y+Z) + s\mu, \qquad \lambda = N/(VW). \end{gathered} \tag{34}$$

The entire source is reconstructed by

$$\begin{gathered} t = A/b, \quad E = C/b, \quad X = (b - 1 - v)/q, \quad k = \lambda b, \\ g = s/V, \quad j = (\lambda A - p)/Y, \quad a = b(R - Y + \Gamma h)/A, \quad w = -r. \end{gathered} \tag{35}$$

Indeed the tail contacts give $tb = A$, $Eb = C$, $b = 1 + v + Xq$; the first column gives $Vg = s$, $tk = p + Yj$, $Ek + Zj + Tg = 1$. Eliminating these shared variables yields (35).

The converse needs the full domain, including strict qualifications. With $R > 1$, define the fixed-coordinate predicate BASE by

$$\begin{gathered} 0 < \beta < 1, \quad 0 < V < 1, \quad 0 < T, Y, Z \leq 1, \\ 1 < p < 1 + \beta, \quad F + \beta < 5, \quad 0 < v \leq s, \\ \Gamma, \nu, \mu > 0, \quad 0 < g \leq 1, \quad \lambda \geq 1, \\ 0 < j < 1 - \beta g, \quad pY - V \leq 1, \\ D_h := A\nu - C\Gamma = \mu + R(\nu - \Gamma) > 0, \\ n_h := A(R - Z) + C(R - Y), \qquad 0 < h_0 := n_h/D_h \leq H := s/\beta. \end{gathered} \tag{36}$$

It implies $q > 0$, $A > C > 0$, and positivity of the reconstruction denominators whenever $b \geq A$. Put

$$\begin{gathered} B_0 = 2 + (1-\beta)v, \qquad B_\tau = 1 + v + qY/V, \\ \sigma_0 = \lambda q - j = \frac{(V+T)(gR - v)}{W} > 0, \qquad B_F = \frac{pq - j(1+v)}{\sigma_0}, \\ I = [A, \infty) \cap (1+v, \infty) \cap (-\infty, B_0] \cap (-\infty, B_\tau) \cap (-\infty, B_F). \end{gathered} \tag{37}$$

These five conditions are exactly $t \le 1$, $X > 0$, $X \le 1$, $\tau > 0$, and $S_{00} < p$. The interval may be empty or a singleton; closed endpoints are retained, and all displayed open endpoints are excluded. Finally define

$$\begin{aligned} \mathfrak{a} &= R(1+q) + Vv, \qquad \mathfrak{c} = V + \beta R, \qquad L(h) = \frac{rA + C(R - Y + \Gamma h)}{A\lambda}, \\ U_O(h) &= \frac{A[B_0 + (1+\beta)h]}{\mathfrak{a} + \mathfrak{c}h}, \qquad U_S(h) = \frac{A[pq + (1+v)(1+\beta h)]}{\mathfrak{a} + \mathfrak{c}h}. \end{aligned} \tag{38}$$

Proposition 6.6 (Complete scale fibre)**.** *Under* BASE*, a complete representative with the specified fixed coordinates and target height exists if and only if*

$$h_0 \le h \le H, \qquad b \in I, \qquad L(h) \le b \le \min\{U_O(h), U_S(h)\} \tag{39}$$

has a solution (b, h). Every solution reconstructs one matrix satisfying all physical bands and all six positive cofactors, with signed pivots $1, p, k, r, -2r$.

Proof. The reconstructed entries satisfy the exact identities

$$\begin{aligned} A(1 - O_{21}) &= D_h h - n_h, \\ Aq(1 + O_{01}) &= A[B_0 + (1+\beta)h] - b(\mathfrak{a} + \mathfrak{c}h), \\ Aq(p + S_{01}) &= A[pq + (1+v)(1+\beta h)] - b(\mathfrak{a} + \mathfrak{c}h). \end{aligned} \tag{40}$$

In addition, $r + Ea \le k$ is $L(h) \le b$, and the receiver bound $|-1-\beta h| \le p$ is $h \le H$. Together with (37), these prove necessity.

For sufficiency, $b \ge A > C > 0$ gives $0 < t \le 1$ and $0 < E < 1$. Membership in I gives $0 < X \le 1$, $\tau > 0$, and $S_{00} < p$. The other five cofactors are positive because $\Gamma, \nu, \mu > 0$, $\delta = 1 - \beta X > 0$, and $\Delta = Z - TX \ge Z - T > 0$. We check both sides of the entry bands, using the same reconstructed entries throughout.

The head has $p - \beta \in (0, 1)$, $L_2 = 1$, $L_0 = pX - 1 \in (-1, 1)$, and $L_1 = pY - V \in (-1, 1]$. The first receiver column has $0 < g \le 1$, $0 < j < 1 - \beta g < 1 < p$, and $P_0 = -\beta g - j \in (-1, 0)$. Its contacts give $S_{10} = p$, $O_{10} = 1$, $O_{20} = -1$, and $S_{20} = Tg - 1 \in [-1, 0]$. Since $k \ge r + Ea > r > 2$ and $Xj < 1$, $S_{00} = k - Xj > 1$. Thus $0 < O_{00} = S_{00} - g < p - g < 1$, since $g = s/V > s$.

In the middle receiver column, $0 < h \le H < 1$, $-p \le \widehat{q}_1 < -1$, and $P_1 = -1$. Equation (40) proves its nontrivial lower bounds. The opposite bounds follow from $S_{01} = -a - X(1+\beta h) < 0$, $O_{01} = S_{01} + h < 1$, $S_{11} = 1 - Vh \in (0, 1)$, and $O_{11} = 1$. Also $S_{21} = r + Ea - Z(1 + \beta h) > r - 2 > 0$, while $O_{21} = S_{21} + Th \le 1$, giving both remaining middle bands. The tail has $P_2 = 1$, $(O_{02}, O_{12}, O_{22}) = (-1, 1, -1)$, and $(S_{02}, S_{12}, S_{22}) = (-1 - v, 1 - Vv, -1 - Tv)$; $v \le s = p - 1$ gives all three p-bands.

For the core, $a > 0$ follows from $R > 1 \ge Y$, and $a < p$ from $S_{01} \ge -p$. Thus $a < p < 2 < k$, $b \le k$, $tk \le k$, $Ek < k$, and $r + Ea \le k$. The remaining entries are $D_{11} = 1 + Y - \Gamma h \in (0, 2)$, $D_{12} = 1 - Yq - Vv \in (-1, 1)$, and $D_{22} = -r + C \in (-r, 0)$. This establishes every original band. The strict first-column capacity comparisons of FP, (B.29) follow from the first-column residual identities and $S_{00} < p$, $P_0 > -1$. □

The equivalence uses I, not its closure. All divisions above are by positive quantities $V, Y, q, A, \lambda, D_h, \sigma_0$, and $\mathfrak{a} + \mathfrak{c}h$ in their declared domains. An unqualified zero denominator is not an infeasibility certificate.

Set

$$f_0 = \gamma = 4132517/10^6, \quad f_1 = 103313/25000, \quad R_i = f_i/2 - 1, \qquad f_0 < \alpha < f_1.$$

The necessary physical inequalities at height f_0 have a complete rational localization certificate in the six coordinates (β, V, s, Y, Z, v). Its conclusion is containment in

$$\begin{aligned} Q = {} & [.617, .6181] \times [.7788, .7795] \times [.4525, .4540] \\ & \times [.97352, .97361] \times [.99998, 1] \times [.1944, .1950]. \end{aligned} \tag{41}$$

All decimals denote exact rationals. The 35 unconditional polynomial rows, one guarded row and their source derivations are in FP, § B.9; the guarded row is used only after its guard is proved on the entire current box. At height f_1, the same necessary system is excluded in Q with its last upper endpoint enlarged to .19501 (FP, Proposition B.24). These are continuous-domain statements, established by covered splits and exact contraction and contradiction rules, not by sampling.

A fixed-target certificate alone would not bound all heights. At the same old head the tail identities give

$$v(R) = \frac{R + Y - t(1 + X)}{t(1 - \beta X) - \Gamma}.$$

The denominator exceeds $1/5$ throughout the height-f_0 localized domain. Thus increasing the target to f_1 changes v by at most $5(R_1 - R_0) = 3/400000$, while β, V, s, Y, Z remain fixed. A source reaching f_1 would therefore yield a representative inside the excluded upper-target box. Consequently every high representative has $F < f_1$. The common bounds from (39) then place it in

$$\begin{aligned} \mathcal{B} = Q_{\beta,V,s,Y,Z} &\times [.1944, .19501]_v \times [.4513, .4566]_h \\ &\times [2.0741, 2.0748]_b \times [R_0, R_1]_R. \end{aligned} \tag{42}$$

The following uniform qualifications make the physical domain explicit before compactness or local inversion is invoked.

Proposition 6.7 (Uniform qualifications)**.** *Throughout $\mathcal{B}$, the reconstruction denominators are positive and*

$$\begin{gathered} q, \Gamma, \nu, \mu, A, C, W, N, D_h, j, g, T > 0, \\ T < 1, \quad g < 1, \quad E < 1, \quad 0 < v < s < \beta < 1, \quad h < H, \\ 1 - j - \beta g > 1/1000, \qquad p - S_{00} > 1/200, \\ pY - V < 1, \quad F + \beta < 5, \quad X > 0, \quad b > 1 + v, \quad 0 < \beta, V, Y < 1. \end{gathered}$$

At points with $X \le 1$, also

$$\delta \ge 1 - \beta > 0, \qquad \tau \ge Y - V > 0, \qquad \Delta \ge Z - T > 0.$$

These are consequences of the 42 exact rational interval checks in FP, Proposition B.25, which also enclose the height-dependent bounds used in (42) and the transport denominator. The last three cofactor bounds are conditional on $X \le 1$; they are not asserted on the whole outer box. In particular, localization has not closed an excluded strict endpoint.

6.3. Physical slacks force the exact height

The eight remaining constraints are the physical slacks

$$\begin{aligned} &G_1 = b - A, && G_2 = B_0 - b, \quad G_3 = \lambda - 1, \\ &G_4 = \lambda b - r - C(R - Y + \Gamma h)/A, \\ &G_5 = A[B_0 + (1 + \beta)h] - b(\mathfrak{a} + \mathfrak{c}h), \\ &G_6 = A[pq + (1 + v)(1 + \beta h)] - b(\mathfrak{a} + \mathfrak{c}h), \\ &G_7 = D_h h - n_h, && G_8 = 1 - Z. \end{aligned} \tag{43}$$

Their feasible directions $G_i \ge 0$ are, respectively, $t \le 1$, $X \le 1$, $b \le k$, $r + Ea \le k$, $O_{01} \ge -1$, $S_{01} \ge -p$, $O_{21} \le 1$, and $Z \le 1$. The factors converting entry bounds into slacks are positive: $G_1 = b(1 - t)$, $G_2 = q(1 - X)$, $G_3 = (k - b)/b$, $G_5 = Aq(1 + O_{01})$, $G_6 = Aq(p + S_{01})$, and $G_7 = A(1 - O_{21})$.

Lemma 6.8 (Exact feasible set)**.** *Let $\mathcal{C}$ be the set of complete scale representatives at heights $F \geq f_0$, with the source and qualifications of Propositions 6.5 and 6.6. Then*

$$\mathcal{C} = \{z \in \mathcal{B} : G_i(z) \geq 0,\ 1 \leq i \leq 8\}. \tag{44}$$

This set is compact, and every member has $F < f_1$.

Proof. Localization, target transport, and the interpretation of the slacks prove the forward inclusion. Conversely, Proposition 6.7 supplies the strict constant conditions in BASE, and $G_3 \geq 0$ supplies $\lambda \geq 1$. Since $R > 1$, $Y < 1$, and $Z \leq 1$, we have $n_h > 0$; hence $G_7 \geq 0$, $D_h > 0$, and $h < H$ give $0 < h_0 \leq h < H$. Thus all of BASE holds. The first two slacks give $A \leq b \leq B_0$. The other endpoints of I follow from $X > 0$, $\tau > 0$, and $p - S_{00} > 1/200$; here the conditional bound for τ uses $G_2 \geq 0$. Finally $G_4, G_5, G_6 \geq 0$ are exactly the lower and upper capacity bounds in (39). Proposition 6.6 reconstructs every band of one complete matrix; the inherited necessary high-value bounds therefore apply to this matrix as well.

All eight slacks are continuous on a neighborhood of $\mathcal{B}$, since their denominators have uniform positive margins. The right side of (44) is consequently closed in a compact box. The strict cofactor and first-column conditions were retained with margins, and the independent upper-target exclusion gives $F < f_1$. □

Theorem 6.9 (Canonical exact height)**.** *Every complete source in $\mathcal{K}$ has height at most α.*

Proof. Suppose a source has height above α. Choose an attainable target $\alpha < f < 5$; Proposition 6.5 then supplies a supercritical member of $\mathcal{C}$. Its maximum has height in (α, f_1).

At fixed s, the Jacobian of the eight slacks with respect to $(\beta, V, Y, Z, v, h, b, R)$ is invertible throughout $\mathcal{B}$. The uniform rational certificate in FP, Proposition B.27 proves

$$\partial R/\partial G_i < -1/250 \qquad (1 \leq i \leq 8).$$

Specifically it bounds $\|I - C_0^T J^T\|_\infty < 7/50$ for a fixed rational preconditioner and encloses the solution of $J^T y = e_R$ in a rational box with all coordinates below $-1/250$. These are whole-box bounds. If a maximum had $G_i > 0$, the inverse-function theorem would allow a short decrease of that slack with the other seven and s fixed. All physical inequalities would remain valid and height would increase. The strict qualifications persist in an open neighborhood; any apparent crossing of an artificial face of $\mathcal{B}$ is excluded by the already proved global localization of every such feasible point. Hence all eight slacks vanish at a maximum.

The equalities force

$$t = X = Z = 1, \quad T = h = s, \quad b = k = r + Ea, \quad O_{01} = -1, \quad S_{01} = -p, \quad O_{21} = 1.$$

In particular $G_5 - G_6 = Aq(h - s)$, so the contact family is forced before stationarity is used. Put $\epsilon = r - 2 = R - 1$ and $\mathcal{L} = (2 - s)(1 + s)$. The top and bottom middle contacts give $a = (1 - \beta)s$ and $\epsilon + Ea = s(\beta - s)$. Hence $b = 2 + s(\beta - s)$, while the tail gives $(1 - \beta)v = s(\beta - s)$ and $Eb = \epsilon + (\beta - s)v$. Substitution in $(b - r)b = a(Eb)$, clearing only $1 - \beta > 0$, yields $2s(\beta - s) = \epsilon\mathcal{L}$. Thus

$$\begin{gathered} \beta = s + \frac{\epsilon\mathcal{L}}{2s}, \qquad v = \frac{s(\beta - s)}{1 - \beta}, \qquad b = 2 + s(\beta - s), \qquad a = (1 - \beta)s, \\ Y = 1 - \frac{\epsilon(s - v)}{s + v}, \qquad V = 1 - \frac{\epsilon[2 + \beta(s - v)]}{s + v}. \end{gathered} \tag{45}$$

The last two formulas solve $R - a = Y(1 + \beta s) - Vs$, $b - R = Yq + Vv$, whose determinant is $v(1 + \beta s) + sq = s + v > 0$. The remaining first-column relation is

$$\epsilon + (\beta - s)v + \frac{b - 1 - s}{Y} + \frac{s^2}{V} = 1. \tag{46}$$

For a compact exact polynomial, define

$$\begin{aligned} d_* &= 2s(1-s) - \epsilon\mathcal{L}, \qquad B_* = 2s^2 + \epsilon\mathcal{L}, \\ Y_n &= s(1-s)(1-\epsilon) + \epsilon^2\mathcal{L}, \qquad Y_d = s(1-s), \\ V_n &= 2s^2(1-s) - 2\epsilon d_* - \epsilon B_* s(1-s) + \epsilon^2 B_*\mathcal{L}, \qquad V_d = 2s^2(1-s). \end{aligned}$$

Then $Y = Y_n/Y_d$, $V = V_n/V_d$, and

$$\begin{aligned} G(s,\epsilon) = [2d_*(1-\epsilon) - \epsilon^2\mathcal{L}^2]Y_nV_n \\ - d_*(2 - 2s + \epsilon\mathcal{L})Y_dV_n - 4d_*s^4(1-s)Y_n. \end{aligned} \tag{47}$$

This integer polynomial has degrees 10 in s and 7 in ϵ. Its exact denominator-clearing identity is

$$\epsilon + (\beta - s)v + \frac{b-1-s}{Y} + \frac{s^2}{V} - 1 = -\frac{G(s,\epsilon)}{2d_*Y_nV_n}. \tag{48}$$

Since $0 < s < \beta < 1$, $d_* = 2s(1-\beta) > 0$, $Y_n = YY_d > 0$, and $V_n = VV_d > 0$, equation (46) is exactly equivalent to $G(s,\epsilon) = 0$ on the physical domain; no zero-denominator component is introduced. The same invertible eight-slack Jacobian parametrizes the contact curve by s in a two-sided neighborhood. The strict physical qualifications persist, and the eight zero slacks preserve feasibility under the reconstruction. Global localization again handles artificial box faces; the maximizing height lies strictly between f_0 and f_1. Maximality therefore implies $\epsilon'(s) = 0$, and differentiating $G(s,\epsilon(s)) = 0$ gives $G_s = 0$, without assuming $G_\epsilon \neq 0$. For this polynomial, exact elimination with the normalization of P_{61} in (2) gives

$$\mathrm{Res}_s(G, G_s) = \frac{\epsilon^{34}(\epsilon-2)(\epsilon-1)^2(\epsilon+1)^7(\epsilon^2-2\epsilon+2)^2}{131072}\,P_{61}(4+2\epsilon).$$

This identity over $\mathbb{Q}$ has degree 109 in ϵ; the factor outside P_{61} has degree 48. The exact resultant check is supplied in FP, Proposition B.30. At the maximum, $132517/2000000 < \epsilon < 3313/50000$, so every factor outside P_{61} is nonzero. Thus $P_{61}(F) = 0$. Root uniqueness in Section 2 forces the maximizing height to be α, a contradiction. □

6.4. Positive rooms and complete coverage

The strict positive room $\mathcal{R}_0^+$ retains all physical bands and satisfies

$$p, k, r > 0, \quad |w| \le r, \quad A, B, d \le 0, \quad \beta, U, V, T, X, Y, Z > 0, \quad \delta, \Gamma, \nu, \tau, \Delta, \mu > 0. \tag{49}$$

The weak room $\mathcal{R}_0^{\ge 0}$ replaces only the last thirteen strict signs by nonnegativity. Physical ties are retained in both rooms.

Proposition 6.10 (Room bounds)**.** *Every strict-room source of height at least H_0 has a same-height complete representative in $\mathcal{K}$. Every weak-room source is a limit of strict-room sources with convergent heights. Consequently both rooms satisfy $F \le \alpha$.*

Proof. Two-face normalization reaches either $e = 1, P_1 = -1$, or $e = 1, P_2 = 1, v_2 \le 0$; the latter face has height below H_0. On the first face the physical shear

$$\begin{aligned} p' = p + \lambda, \quad \beta' = \beta + \lambda, \quad u' = u + \lambda x, \quad q' = q - \lambda v, \\ \lambda = \min\{1-\beta, (1-U)/X, (1-V)/Y, (1-T)/Z\}. \end{aligned}$$

preserves all bands, cofactors, D and height. The identity

$$1 + U + (1-\beta)X - k = (1 - O_{00}) + X(1 + P_0) + \delta(1 + v_0) \ge 0$$

and $k \geq r > 2$ imply $\beta' < 1$; positivity of ν' gives $T' < 1$. Thus $U' = 1$ or $V' = 1$, including tied minima and zero shear. The latter face has height below H_0, so the endpoint belongs to $\mathcal{K}$, with $t = c, E = -d, a = -A, b = -B$. The face bounds and full source qualifications are recorded in FP, Proposition E.2.

For the weak room, we give the physical approximation explicitly. For $0 < \eta < 1/3$, put

$$H_\eta = \mathrm{diag}(1, 1-\eta, 1-2\eta, c_\eta, c_\eta), \qquad c_\eta = 1 - 3\eta, \quad M_\eta = H_\eta M H_\eta.$$

Positive diagonal scaling commutes with Schur elimination in the sense that, writing $H = \mathrm{diag}(h_1, \ldots, h_5)$, every surviving (i, ℓ) entry is multiplied by $h_i h_\ell$:

$$(HMH)_{i\ell} - \frac{(HMH)_{ij}(HMH)_{j\ell}}{(HMH)_{jj}} = h_i h_\ell \left(M_{i\ell} - \frac{M_{ij} M_{j\ell}}{M_{jj}} \right).$$

For the first three pivots, every competing entry has $h_i h_\ell < h_j^2$; their complete-pivoting bounds therefore become strict. The equal last two factors preserve the normalized X-tail, with height $F_\eta = c_\eta^2 F$. Each weak cofactor is multiplied by a strictly positive factor, and the signs of A, B, d are unchanged. The receiver upper bounds and the physical bands affected below have strict margins.

Fix η and suppress primes. Consider $n_0 = (U, X), n_1 = (V, Y), n_2 = (\beta, 1), n_3 = (T, Z)$. For each nonzero n_i, write $n_i = m_i(t_i, 1-t_i)$, where $m_i > 0$. All six weak cofactors give $t_i \geq t_j$ whenever $i < j$ and both vectors are nonzero. At a zero vector set $m_i = 0$ and choose t_i by monotone interpolation between its nearest nonzero neighbors, using endpoint values 1 and 0 when needed. Thus $1 \geq t_0 \geq t_1 \geq t_2 \geq t_3 \geq 0$. For $0 < \varepsilon < 1$, define

$$t_i^\varepsilon = (1-\varepsilon) t_i + \varepsilon(4-i)/5,$$
$$n_i^\varepsilon = (m_i + \varepsilon)(t_i^\varepsilon, 1 - t_i^\varepsilon) \quad (i = 0, 1, 3), \qquad n_2^\varepsilon = \left(\frac{t_2^\varepsilon}{1 - t_2^\varepsilon}, 1 \right).$$

Now $1 > t_0^\varepsilon > t_1^\varepsilon > t_2^\varepsilon > t_3^\varepsilon > 0$, and $\det(m(t, 1-t), n(s, 1-s)) = mn(t-s)$ makes all six cofactors strictly positive. The new vectors converge to the contracted ones, including the zero vectors. Since the contracted $\beta < 1$, both $t_2 < 1/2$ and its target $2/5 < 1/2$, so the new β also remains below one.

Keep the contracted p, D, e, v, q fixed and reconstruct the one matrix (7) with these new β, u, x. For sufficiently small ε, the strict margins preserve every affected physical band; D, its tail geometry and its height remain unchanged. This is a strict-room source of height $c_\eta^2 F$, so the preceding bound gives $c_\eta^2 F \leq \alpha$. For each η, choose an admissible $0 < \varepsilon(\eta) \leq \eta$ and let $\eta \downarrow 0$. The matrices converge to the original weak source and $F \leq \alpha$ follows. All six weak cofactors are needed to order the nonzero rays across zero vectors. This argument does not assert a same-height strict representative for a weak source. □

A qualified local wall entrance handles near-peak sources not already in a room. In the ordered native coordinates

$$z = (k, r, w, A, B, c, d, p, e, \beta, u_0, u_1, u_2, x_0, x_1, x_2, v_0, v_1, v_2, q_0, q_1, q_2),$$

let $\bar{c}_j$, $0 \leq j \leq 3$, be the first 22 coordinates of the rational centres in FP, § C.2, using the declared order of (63).

Lemma 6.11 (Qualified wall entrance)**.** *A complete X source satisfying*

$$\|z - \bar{c}_j\|_\infty + 3(r + w) < 1/1250 \tag{50}$$

has a complete physical path to $r + w = 0$, followed by a signed tail permutation, whose strict-room endpoint has height at least $F + (r + w)/10$. Thus $F \leq \alpha - (r + w)/10$.

Proof. The 22-coordinate wall certificate fixes p and twenty physical slacks, decreases $r + w$ at unit speed, and bounds speed by 3 and height gain below by $1/10$. All other bands have positive margins throughout the radius-$1/1250$ box. Condition (50) therefore keeps the flow inside this box until the wall. The signed permutation there gives all strict signs in (49). When $r + w = 0$, the path has zero length. The exact flow and sign certificate is identified in FP, Lemma E.3; Proposition 6.10 then bounds the endpoint. □

Proof of Proposition 6.1. The complete source certificates have the following overlapping cover:

Source domain	Conclusion
$0 < k \le 2$	$F < \Lambda$
$k \ge 21/10$	$F < \gamma$
$2 < k \le 21/10,\ r > k$	$F < \gamma$
$\gamma/2 \le k \le 21/10,\ r \le k,\ F \ge \gamma$	$F \le \alpha$

The first three domains leave $2 < k < 21/10, r \le k$. If $F \ge \gamma$, then $\gamma \le F \le 2r \le 2k$, placing the source in the last domain. Its two accepted roots use exact contradiction leaves, direct rational height bounds, weak-room exits, and the qualified wall entrance. All branches close under the source-preserving acceptance rules of Section 8. The original-root domain maps, finite payloads and complete acceptance records are in Supplementary Index S4 and FP, § E. The interfaces $k = 2, 21/10$ and $r = k$, other signs, zero receivers and cofactors, and physical ties remain covered. A safe leaf may contain feasible points; it need not exclude $F \ge \gamma$. □

The order is canonical height, room bounds, qualified exits, then the complete X cover. The later 24-coordinate two-gap transport uses this complete bound at its endpoint; it is not a premise of it.

7. Exact reduction of the negative diagonal domain

Replace (19) by

$$D = \begin{pmatrix} k & A & B \\ ck & r + cA & s + cB \\ dk & t + dA & -r + dB \end{pmatrix}, \qquad 0 \le s, t \le r. \tag{51}$$

The safe boundary is included. A complete source satisfies (7), all bands (9)–(12), $\|D\|_{\max} \le k$, and these tail conditions. Since the first Schur tail is a real completely pivoted 4×4 matrix,

$$F \le 4p. \tag{52}$$

Thus $F > 4$ implies $p > 1$ and $e\beta \ge p - 1 > 0$. Simultaneous sign changes of the second row and column, of the remaining three rows and columns, and of their last two rows and columns permit, respectively, $e, \beta > 0$, $x_0 \ge 0$, and $A \le 0$, without changing height or the tail conditions. Other receivers and c, d retain both signs.

Fix the seventeen shared coordinates

$$\vartheta = (k, A, B, c, d, u_0, u_1, u_2, x_0, x_1, x_2, v_0, v_1, v_2, q_0, q_1, q_2). \tag{53}$$

We maximize the six actual variables (β, p, e, r, s, t) by three feasible paths. The entry bounds throughout refer to one reconstructed matrix; the capacities below are simultaneously attained.

7.1. Prefix maximization

Let

$$I_\beta = [0,1] \cap \bigcap_{j=0}^{2} \{b : |q_j + bv_j| \le 1\}, \qquad \bar\beta = \max I_\beta.$$

An empty interval excludes the fibre. A zero v_j is handled by $|q_j| \le 1$, before division.

Lemma 7.1. *For a complete source with $p \ge 1$ and $0 \le e, \beta \le 1$, the segment increasing β to $\bar\beta$, with the other variables fixed, preserves complete feasibility and height.*

Proof. The prefix interval is convex; $p - e\beta \le 1$ improves, its lower bound is automatic from $p \ge 1$ and $e\beta \le 1$, and no other band or the height depends on β. □

If $\bar\beta = 0$, the head forces $p = 1$, safe by (52). Otherwise form the head row $(a_0, b_0) = (1, \bar\beta)$, and, for each $x_i \ne 0$, the row $(a_i, b_i) = (|x_i|, \operatorname{sgn}(x_i)u_i)$. There are at most four rows, $0 < a_i \le 1$, $|b_i| \le 1$. Zero x_i give automatic bands. The remaining prefix polygon is exactly

$$a_i p - b_i e \le 1, \qquad 0 \le e \le 1, \quad p \ge 1. \tag{54}$$

The reverse bounds are automatic. Define

$$\begin{aligned} P &= \min\left\{ \frac{1+b_i}{a_i}\ (b_i \ge 0), \quad \frac{b_i - b_j}{a_j b_i - a_i b_j}\ (b_i > 0 > b_j) \right\}, \\ e_* &= \max\left(0, \max_{b_i > 0} \frac{a_i P - 1}{b_i}\right). \end{aligned} \tag{55}$$

All displayed denominators are positive.

Lemma 7.2. *The maximum feasible p is P, attained at (P, e_*), with $1 \le P \le 2$. At most six capacities occur in* (55).

Proof. For $\ell_i = 1 - a_i p + b_i e \ge 0$, the identities

$$1 + b_i - a_i p = \ell_i + b_i(1 - e), \qquad b_i - b_j - (a_j b_i - a_i b_j)p = b_i \ell_j - b_j \ell_i$$

prove necessity under the stated sign conditions. Conversely, positive rows give lower bounds for e, negative rows upper bounds, and zero rows only bounds on p. The capacities compare every lower bound with one and with every upper bound. Pairing a negative row with the head also gives

$$\frac{\bar\beta - b_j}{a_j \bar\beta - b_j} \le \frac{1}{a_j},$$

so each upper bound is nonnegative. Hence e_* satisfies all rows. Each capacity is at least one, and the head gives $P \le 1 + \bar\beta \le 2$. If there are m, z, n positive, zero and negative rows, the count is $m + z + mn \le 6$, since $m \ge 1$ and $m + z + n \le 4$. □

The segment to (P, e_*) stays in the prefix polygon. Along it p increases, while D, $S = D + xq^T$, and $O = D + xq^T + uv^T$ stay fixed. The stage bounds $|q_j|, |S_{ij}| \le p$ improve; all other bands and the height are preserved.

7.2. Tail maximization and the physical root

At $p = P$, set

$$\begin{aligned} \ell_{ij} &= \max(-k, -P - x_i q_j, -1 - x_i q_j - u_i v_j), \\ U_{ij} &= \min(k, P - x_i q_j, 1 - x_i q_j - u_i v_j). \end{aligned} \tag{56}$$

Check the five fixed entries in the zeroth row or column of D and $|q_j| \le P$; failure cannot be repaired by decreasing p. For the remaining entries define

$$\begin{aligned} r_- &= \max(0, \ell_{11} - cA, dB - U_{22}), & R &= \min(U_{11} - cA, dB - \ell_{22}), \\ s_- &= \max(0, \ell_{12} - cB), & s_+ &= U_{12} - cB, \\ t_- &= \max(0, \ell_{21} - dA), & t_+ &= U_{21} - dA, \\ S_* &= \min(R, s_+), & T_* &= \min(R, t_+). \end{aligned}$$

Theorem 7.3 (Attained six-variable maximum)**.** *A nonempty high-value fibre has the complete feasible endpoint $(\beta, p, e, r, s, t) = (\bar{\beta}, P, e_*, R, S_*, T_*)$, and*

$$\mathcal{B}(\vartheta) = R + \frac{S_* T_*}{R} = \max_{\text{complete six-variable fibre}} F. \tag{57}$$

After the prefix and fixed-entry checks, the positive-tail fibre is nonempty exactly when $R > 0$, $R \ge r_-$, $S_* \ge s_-$, and $T_* \ge t_-$. Every source reaches the endpoint by three complete feasible paths with nondecreasing height.

Proof. The first two paths were proved above. Their endpoint has the exact tail feasible set

$$r_- \le r \le R, \quad s_- \le s \le s_+, \quad t_- \le t \le t_+, \quad r > 0, \quad 0 \le s, t \le r.$$

Nonemptiness implies $R \ge r > 0$, $S_* \ge s$, $T_* \ge t$; the stated conditions put (R, S_*, T_*) in this same set. For every feasible tail,

$$Rr(\mathcal{B} - F) = (R - r)(Rr - st) + rt(S_* - s) + rS_*(T_* - t) \ge 0, \tag{58}$$

because $st \le r^2$ and all other factors have the indicated signs. The segment to the endpoint remains feasible and increases each coordinate. Its height is nondecreasing since $\partial_r F = 1 - st/r^2 \ge 0$, $\partial_s F = t/r \ge 0$, $\partial_t F = s/r \ge 0$. □

Every hypothetical $F > \alpha$ source has its framework in the closed root Q_0 defined by

$$\begin{gathered} 4\gamma/9 \le k \le 9/4, \quad -9/4 \le A \le 0, \quad |B| \le 9/4, \quad |c|, |d| \le 1, \\ |u_i|, |v_i| \le 1, \quad 0 \le x_0 \le 1, \quad |x_1|, |x_2| \le 1, \quad |q_i| \le 2. \end{gathered} \tag{59}$$

Here $F \le (9/4)k$ gives the lower endpoint for k; the other bounds follow from the complete bands. In particular $k \le 2$ is retained. The additional necessary budget $pk \le 4$, when used, follows from the leading 3×3 determinant: its absolute value is pk, and its maximum on the unit-entry cube is four. Indeed multilinearity reduces this maximum to sign matrices, whose determinants are multiples of four, whereas $3\sqrt{3} < 8$.

Definition 7.4 (Complete fibre and high framework)**.** Let $\mathcal{F}(\vartheta)$ consist of the tuples $\zeta = (\beta, p, e, r, s, t)$ with $p \ge 1$, $k, r > 0$, $0 \le e, \beta \le 1$, $0 \le s, t \le r$, for which (7), (51), all complete bands, and $\|D\|_{\max} \le k$ hold for the same reconstructed matrix. Set

$$\mathcal{H}_\gamma = \{\vartheta \in Q_0 : \exists \zeta \in \mathcal{F}(\vartheta), \quad r + st/r \ge \gamma\}.$$

Proposition 7.5 (Root semantics)**.** *Every proper negative-diagonal source with $F > \alpha$ has a height-preserving sign normalization with framework in $\mathcal{H}_\gamma$. For every $\vartheta \in \mathcal{H}_\gamma$, Theorem 7.3 realizes $\mathcal{B}(\vartheta)$ by a member of $\mathcal{F}(\vartheta)$. It therefore suffices to prove $\mathcal{B} \le \alpha$ on $\mathcal{H}_\gamma$.*

Proof. Use (52), $\alpha > 4$, the actual sign changes, the universal root bounds, and $F > \alpha > \gamma$; the maximum is attained by Theorem 7.3. □

All intersections and selectors include ties. Zero coefficients are handled before division, and $R = 0$ cannot support a high source. A rational box alone is not a physical source: necessary relaxations must contain each complete source with all shared products assigned their actual values. A safe transformation must moreover reconstruct a complete physical image of no smaller height.

8. Rational certificates and mixed-height coverage

Discovery proposes subdivisions and certificates; acceptance checks their declared sources by exact rational arithmetic. Floating-point search is not an acceptance rule. Child boxes cover the parent; an open terminal has not yet proved exclusion above γ or safety at α. An attachment discharges it only if its entire inherited source agrees.

8.1. Necessary inequalities and exact terminal rules

For $x \in [L_x, U_x]$, $y \in [L_y, U_y]$, one shared value $z = xy$ satisfies

$$z \geq L_x y + L_y x - L_x L_y, \qquad z \geq U_x y + U_y x - U_x U_y,$$
$$z \leq L_x y + U_y x - L_x U_y, \qquad z \leq U_x y + L_y x - U_x L_y.$$

Every complete source supplies all products simultaneously. Enclosures are outward; divisions and capacity selectors require signs valid over the whole region or subdivision retaining every alternative. If the necessary rows are $a_j \cdot z \leq b_j$, exact nonnegative weights give

$$a = \sum_j \lambda_j a_j, \quad B = \sum_j \lambda_j b_j, \qquad a \cdot z \leq B, \quad \lambda_j \geq 0.$$

For coordinate bounds $[L_i, U_i]$, the acceptor checks

$$m = \sum_i \min(a_i L_i, a_i U_i) \leq a \cdot z. \tag{60}$$

The strict rational comparison $m > B$ excludes every real source in the region; it does not assume rational matrix entries. An interval contradiction is the one-expression case. The actual height may be introduced by both directions of $rF - r^2 - st = 0$, never by the X-specific objective $r - w$.

8.2. A real source-bound terminal

One terminal of the final negative-diagonal certificate excludes a canonical high source $F \geq \gamma$ with all ancestor conditions retained; it does not exclude the entire geometric box. It uses 24 source coordinates, 43 shared products, and 692 necessary rows. Its 49 selected rows have positive integer weights; for example,

row	inequality	weight
2	$-z_0 - z_4 \leq 0$	31 317 836
9	$-z_0 + z_2 + z_{36} \leq 0$	229 060 761

$z_{36} = z_5 z_6$.

Indices are zero-based. All 49 rows, not just these two, give the exact rational margin

$$m - B > 1846 > 0. \tag{61}$$

The exact fraction and its calculation are given in FP, § F; FP, § D.3 locates the full row dictionary, coordinate bounds and weights. The inherited parent, projection and transformation prefix establish their applicability; the positive margin alone does not. This terminal is one position of the complete 54-terminal local cover described below. Its margin depends on the weight normalization and is not a bound on $\alpha - F$.

Safety without exclusion. If $F = h \cdot z$, take $t > 0$, $c = a - th$ and $m_c = \sum_i \min(c_i L_i, c_i U_i)$. Then

$$tF = a \cdot z - c \cdot z \le B - m_c. \tag{62}$$

Thus $B - m_c \le t\alpha_-$ proves $F \le \alpha$, not infeasibility. Alternatively, an actual complete image $T(z)$ with $F(z) \le F(T(z)) \le \alpha$ proves safety. The following transport provides such an image near cross-saturated tails.

8.3. Uniform transport to the complete cross-saturated domain

Use the complete coordinates

$$\begin{aligned}\xi = (&k, r, w, A, B, c, d, p, e, \beta, u_0, u_1, u_2, x_0, x_1, x_2,\\ &v_0, v_1, v_2, q_0, q_1, q_2, \sigma, \tau),\end{aligned} \tag{63}$$

where $\sigma = r - s$, $\tau = r - t$, and

$$H = \begin{pmatrix} r & r-\sigma \\ r-\tau & w \end{pmatrix}, \qquad F = \frac{(r-\sigma)(r-\tau)}{r} - w. \tag{64}$$

The proper negative-diagonal domain has $w = -r$, $\sigma, \tau > 0$. The actual third Schur matrix and its two parent blocks are

$$D = \begin{pmatrix} k & A & B \\ ck & r + cA & r - \sigma + cB \\ dk & r - \tau + dA & w + dB \end{pmatrix}, \qquad S = D + xq^T, \quad O = S + uv^T. \tag{65}$$

Substitution in (7) gives one real matrix $M(\xi)$, whose successive Schur blocks are $\left(\begin{smallmatrix} p & q^T \\ px & S \end{smallmatrix}\right)$, D, and H in (64). These are algebraic identities for $p, k \neq 0$; the last elimination also requires $r \neq 0$. All vector and block indices below are $0, 1, 2$.

For a band $|f| \le b$, write $f^+ = b - f$, $f^- = b + f$. These superscripts denote slacks, not positive and negative parts. The physical bands have the following polynomial slacks:

$$\begin{gathered} e^{\pm} = 1 \mp e, \quad \beta^{\pm} = 1 \mp \beta, \quad \mathrm{head}^{\pm} = 1 \mp (p - e\beta), \\ u_i^{\pm} = 1 \mp u_i, \quad x_i^{\pm} = 1 \mp x_i, \quad v_i^{\pm} = 1 \mp v_i, \quad q_i^{\pm} = p \mp q_i, \\ L_i^{\pm} = 1 \mp (px_i - eu_i), \qquad P_i^{\pm} = 1 \mp (q_i + \beta v_i), \\ D_{ij}^{\pm} = k \mp D_{ij}, \quad S_{ij}^{\pm} = p \mp S_{ij}, \quad O_{ij}^{\pm} = 1 \mp O_{ij}. \end{gathered} \tag{66}$$

Omit the identically zero slack $D_{00}^{+} = k - k$. The resulting 95 polynomials, together with the nine slacks

$$r + w, \quad r - w, \quad p, \quad k, \quad r, \quad \sigma, \quad \tau, \quad r - \sigma, \quad r - \tau, \tag{67}$$

form the complete list of 104 source inequalities. Their nonnegativity, with $p, k, r > 0$, gives a normalized matrix and legal leading complete pivots $1, p, k, r$. Conversely a source with this fixed-order tail satisfies these same bands. Competing entries may tie. On the boxes below $w < 0$, so its last pivot is $-F$, with $F > 0$.

Let y be ξ with p deleted, retaining the order in (63). For each chart $j = 0, 1, 2, 3$, let $\ell_{j,0}, \ldots, \ell_{j,19}$ be the twenty slacks in Table 1, in index order, and define

$$\Phi_{j,p}(y) = (\ell_{j,0}, \ldots, \ell_{j,19}, r + w, \sigma, \tau), \qquad J_j = D_y \Phi_{j,p}. \tag{68}$$

Thus J_j is a 23×23 matrix, polynomial in all 24 coordinates, including the parameter p. The four rational centres c_j are specified in FP, § C.2. Write

$$Q_j = \{\xi : \|\xi - c_j\|_\infty \le R\}, \qquad R = 1/1250. \tag{69}$$

The derivative fixes p, but the estimates hold uniformly for its entire interval in Q_j; they do not restrict the source to a centre's value of p.

Table 1: The twenty selected physical slacks in each chart. The index is within the selected list, not within all 104 source inequalities.

Index	$j=0$	$j=1$	$j=2$	$j=3$	Index	$j=0$	$j=1$	$j=2$	$j=3$
0	e_0^+	e_1^+	e_2^+	e_3^+	10	O_{10}^+	O_{10}^+	O_{10}^-	O_{10}^+
1	u_0^+	u_1^+	u_2^+	u_3^+	11	O_{11}^+	O_{11}^+	D_{11}^+	O_{11}^+
2	x_0^+	x_1^+	x_2^+	x_3^+	12	O_{12}^+	O_{12}^+	O_{11}^+	D_{12}^+
3	S_{01}^-	D_{01}^-	S_{01}^-	D_{01}^-	13	x_2^+	x_2^-	O_{12}^+	O_{12}^+
4	O_{01}^-	O_{01}^-	O_{01}^-	O_{01}^-	14	L_2^+	L_2^-	P_2^-	P_2^+
5	D_{02}^-	S_{02}^-	D_{02}^+	S_{02}^+	15	P_2^+	P_2^-	D_{20}^+	D_{20}^+
6	O_{02}^-	O_{02}^-	O_{02}^+	O_{02}^+	16	O_{20}^-	O_{20}^+	S_{20}^+	S_{20}^+
7	P_1^-	P_1^+	x_1^+	x_1^-	17	D_{21}^+	O_{21}^+	O_{20}^+	O_{20}^+
8	D_{10}^+	D_{10}^+	L_1^+	L_1^-	18	O_{21}^+	D_{22}^-	O_{21}^+	O_{21}^+
9	S_{10}^+	S_{10}^+	P_1^-	P_1^+	19	O_{22}^-	O_{22}^-	O_{22}^-	O_{22}^-

Lemma 8.1 (Local transport)**.** *If ξ_0 is complete, $\sigma_0, \tau_0 \geq 0$, and*

$$d_0 + 3(\sigma_0 + \tau_0) < R, \qquad d_0 = \|\xi_0 - c_j\|_\infty , \tag{70}$$

then a piecewise continuously differentiable path inside Q_j joins it to a complete X source ξ_*, preserving p, $r + w$, and the twenty selected slacks, and satisfies

$$F(\xi_*) - F(\xi_0) \geq (\sigma_0 + \tau_0)/10, \qquad F(\xi_0) \leq \alpha - (\sigma_0 + \tau_0)/10. \tag{71}$$

The selected slacks need not vanish initially.

Proof. The finite rational calculations in FP, § C give rational preconditioners C_j with, throughout Q_j,

$$\|I - C_j J_j\|_\infty \leq \eta = 396725479558446119/10^{19} < 1/25. \tag{72}$$

Thus J_j is invertible. For the two final coordinate vectors, put $v_\sigma = J_j^{-1} e_\sigma$, $v_\tau = J_j^{-1} e_\tau$. The same whole-box checks establish

$$\|v_\sigma\|_\infty , \|v_\tau\|_\infty < 3, \qquad D_y F v_\sigma, D_y F v_\tau \leq -1/10, \tag{73}$$

and the 81 unselected slacks have the common positive lower bound

$$\mu = 913752034886059/10^{18} > 0. \tag{74}$$

Also $r \geq 2065458539/10^9 > 0$, $w \leq -2065458539/10^9$. Consequently the expression for F is the positive final-pivot magnitude of every physical source here. These checks use explicit polynomial coefficients and rational matrices, not numerical trajectories.

Fix $p = p(\xi_0)$. Solve $\dot{y} = -v_\sigma(p, y)$ for time σ_0, then $\dot{y} = -v_\tau(p, y)$ for time τ_0. The equation $J_j \dot{y} = -e_\sigma$, respectively $-e_\tau$, preserves the selected slacks and $r+w$, while decreasing just the chosen gap at unit speed. Zero gaps require no segment. Total displacement is at most $3(\sigma_0+\tau_0)$. By (70), this prevents first exit and keeps the paths in a compact subset of the open set where the vector fields are smooth. The continuation theorem therefore guarantees both prescribed endpoints. The preserved slacks and $r + w$ remain nonnegative; the gaps do not cross zero; the other slacks stay positive by (74). Hence all matrix and stage bounds hold simultaneously along the paths, allowing pivot ties. At the endpoint both gaps vanish, giving a complete X source. Integration of $\dot{F} \geq 1/10$ gives the height gain, and Proposition 6.1 bounds the endpoint. □

The four inner boxes $\|\xi - c_j\|_\infty \leq 1/10000$ satisfy the strict guard for nonnegative gaps, since $7/10000 < 1/1250$. The acceptor requires the guard on the entire source enclosure. Equality at the outer guard is not accepted by continuity; it stays with the sources requiring subdivision or another terminal. These four local boxes are not claimed to cover the whole negative domain.

8.4. Complete composition

Lemma 8.2 (Mixed coverage)**.** *Let $\gamma < \alpha$. Suppose a rational root contains a canonical image, of no smaller height, of every hypothetical source with $F > \alpha$. Suppose a finite acyclic rooted proof graph covers that root by exact subdivisions, qualified contractions and complete coordinate changes of no smaller height. If every terminal either excludes complete sources with $F \geq \gamma$, or proves $F \leq \alpha$, and every child and terminal is bound to its full declared source, no original source has $F > \alpha$.*

Proof. Follow a hypothetical high source through the graph. It belongs to a closed child at every subdivision. A contraction has already certified discarded sources impossible or safe; each coordinate change retains a complete image of no smaller height. Finiteness gives a terminal, contradicting either of its two assertions. Boundary overlaps cause no loss of coverage. □

Attachments must agree in model, parent, root path, endpoints and full transformation prefix, not merely dimension or a success label. Hashes identify the bytes of previously accepted proofs; the acceptance rules and their source inclusions give mathematical validity.

9. The complete negative-diagonal certificate

Proposition 9.1. *Every complete real source satisfying* (51)*, $p, k, r > 0$, $0 < s, t < r$, has $F = r + st/r \leq \alpha$.*

The model uses the whole root (59) and the physical fibre semantics of Theorem 7.3, without imposing a selected tail-contact equation on every canonical image. The Supplementary Index, S5–S7, identifies its model, rules and full proof objects.

Exact verification of the base tree accepted 749693 nodes: 374846 splits, 374839 contradiction terminals, four safe terminals, and four open terminals. Its 750 enclosed continuation certificates had no internal open terminal. Denote the four remaining original-root dyadic boxes by $Q_1, \ldots, Q_4$. Complete certificates cover respectively $111, 240, 159, 56$ constituent sources, each retaining its ancestor transformation prefix.

For Q_2, 239 sources retain accepted component proofs; the last uses a complete prefix-free 54-terminal subtree: 48 contradictions, three height or feasible-transport certificates, and three canonical threshold exclusions. Each of the latter covers both qualified charts of its source. The terminal in Section 8.2 is one of these 54 positions, not a substitute for the other 53.

The local completion is attached to that exact source of Q_2. The four complete parent certificates are then attached to the base tree's four open positions. A traversal reconstructs every open box from the actual original root and checks model, path, endpoints and complete transformation prefix against each attachment. These boxes are exactly $Q_1, \ldots, Q_4$; no original-root source remains uncovered. The 566 constituent sources discharge four original-parent obligations, which together with 465 previously covered parents give 469 of 469. This count is only a consistency check: coverage rests on the exact traversal and source comparisons detailed in FP, § D.

Proof. A hypothetical $F > \alpha$ source has $p > 1$. Its sign normalization and the three paths of Theorem 7.3 give a complete framework in Q_0 with $\mathcal{B} > \alpha$. The accepted base proof and four matching full-parent completions form the complete finite graph above. Lemma 8.2 excludes this source at a contradiction or safe terminal. □

The final composer checks identities and exact source bindings of previously accepted mathematical proofs; it does not rerun every terminal's arithmetic. The base tree's four open records are preserved and discharged by their attachments. Likewise safety at α does not imply infeasibility at γ.

10. Completion of the global proof

Proof of Theorem 1.2. The lower bound is [2, Theorem 2.2], with the exact corresponding witness identified in Section 2. For the upper bound take any nonzero real matrix and any legal pivot path, normalize $\|A\|_{\max} = 1$, and absorb the path's permutations. A zero pivot forces its trailing block to vanish. Corollary 3.2 bounds growth through the fourth stage by $4 < \alpha$. For a larger fifth pivot, Proposition 4.1 and Lemma 5.1 give a same-height complete source on the diagonal- or cross-saturated boundary. Actual sign changes put every cross-saturated source in (16), bounded by Proposition 6.1. For a diagonal-saturated source, the positive-sign or zero-arm case of (17) has height at most $r \leq 4$, and a tied arm belongs to X. The remaining case is exactly (18), bounded by Proposition 9.1. Thus every legal path has growth at most α, proving equality. □

11. Reproducibility and scope

Discovery proposes cuts, subdivisions and transformations; verification checks their mathematical conditions; composition checks that the verified pieces cover the original sources. A hash fixes a file's identity, not its mathematical validity. Fresh reproduction must execute the mathematical producers of the required records before composition; inspecting retained records and their bindings is a record-and-composition audit.

AI-assisted research workflow. Agent systems were used to explore candidate reductions, draft arguments, and develop symbolic, exact-verification and Lean source code. Their candidate outputs were checked through the analytic arguments and exact acceptance rules stated in this paper; model responses alone were never accepted as proof terminals. The tools, their purposes and human oversight are disclosed in the final AI-use statement.

The exact verifier code, certificate data, Lean source and verification instructions are available in the fixed release `rho5-proof`, release `v1.0.1`. The Supplementary Index S1–S8 supplies source identities, commands, recursive dependencies and the scope of recorded runs. It is an index, not the complete data archive. The original preprint and index are archived at Zenodo v1.0.1; the expanded public version is Zenodo v1.0.2.

A fresh negative-diagonal replay on 30 September 2026 executed all 570 planned mathematical producers, covered the 566 distinct target sources, and matched the four completed parent paths and rational boxes to the original root, giving effective open count zero. The original root itself retained its four prescribed open positions, discharged by those attachments. Its sixteen-worker rational replay took about 3.60 hours; this is elapsed time for that root run, not total discovery or end-to-end verification time. This replay reused upstream analytic and complete X inputs. It was neither a new-machine replay nor an independent implementation. FP, § D.4 and Index S7–S8 give the execution order, acceptance conditions and full scope.

The implementations of the exact rational/interval acceptors, their parsers, model generation and source-cover composition are part of the trust boundary. Exact arithmetic removes rounding uncertainty but cannot by itself ensure correct guards or complete coverage. No floating-point optimizer or search heuristic is accepted as a proof terminal. A tested, location-independent driver for the entire transitive dependency chain in one fresh environment is not yet supplied; historical absolute paths need explicit remapping without changing mathematical input identities.

Partial Lean formalization. The companion development formalizes the algebraic constant and its attainment, key analytic reductions, and a conditional final theorem. It retains the two global safety hypotheses `XGlobalSafety` and `RootEndpointSafety`. Their full discharge, including the remaining model and coverage connections, is not claimed. The Python/C++ acceptors are not formally verified. The frozen first-phase project was rebuilt on Linux with Lean 4.30.0 from an empty project-object directory: 634 product modules, three example/audit

modules and 47 named axiom checks passed. Third-party caches were reused. This establishes the recorded build scope, not an unconditional Lean proof of the present theorem.

The theorem determines the value and an attaining matrix. Classification of all maximizers, uniqueness modulo symmetry and the analogous complex value remain separate questions.

AI use and acknowledgments

AI models and agent systems contributed substantially to mathematical exploration, proof drafting, symbolic and exact-arithmetic code, Lean source development, and manuscript revision. Qianli Ma led the research, managed its progress and budget, assigned responsibilities among the agents, and made the final decisions to select, continue or terminate research approaches. The tools included GPT-6 Astra, GPT-6.1 during revision, Claude Code, Wujie AI AGENT and DeepSeek Harness; exact underlying model versions were not retained for every agent system. Their outputs were checked through the stated analytic and exact-verification workflow; AI responses are not proof evidence. The authors retain responsibility for the mathematical claims, code, data and manuscript.

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